Ageing Research Reviews 114 (2026) 102965

Contents lists available at ScienceDirect

Ageing Research Reviews

journal homepage: www.elsevier.com/locate/arr

Review article

Multivitamin and mineral use: A rapid review of meta-analyses on health outcomes

Weilan Wang [a,b,1], Vanessa Kristina Wazny [a,b,1], Muhammad Daniel Azlan Mahadzir [a,b], Andrea Britta Maier [a,b,c,*]

[a] Healthy Longevity Translational Research Programme, Yong Loo Lin School of Medicine, National University of Singapore, Singapore
[b] NUS Academy for Healthy Longevity Yong Loo Lin School of Medicine, National University of Singapore, Singapore
[c] Department of Human Movement Sciences, @AgeAmsterdam, Faculty of Behavioural and Movement Sciences, Vrije Universiteit Amsterdam, Amsterdam Movement Sciences, Amsterdam, the Netherlands

ARTICLE INFO

Keywords:
Vitamins
Minerals
Dietary supplements
Cognition
Neoplasms
Longevity

ABSTRACT

Multivitamin and mineral (MVM) supplements are among the most widely used dietary supplements globally, however, their role in promoting healthspan and longevity remains unclear. This review evaluated comprehensive findings from meta-analyses to clarify their health effects. A rapid review of MEDLINE and EMBASE identified 19 eligible meta-analyses published from 2000 to 2025, encompassing 5535,426 participants, including over 333,943 pregnancies and 904,947 children exposed to maternal MVM supplementation. Randomized controlled trials indicated that MVM use improved global cognition, episodic memory, and immediate recall in older or cognitively intact adults, reduced psychological symptoms in healthy individuals, and lowered systolic blood pressure in at-risk populations. However, no benefits were found for all-cause mortality, COVID-19 outcomes, visual acuity, or multiple cognitive domains, and a higher risk of age-related macular degeneration progression was reported. Observational studies found associations between MVM use and a reduced risk of colorectal cancer, coronary heart disease, cataracts, and fragility hip fractures, but not breast or prostate cancer, stroke, or overall mortality. During pregnancy, MVM supplementation was linked to reduced risks of small-for-gestational-age births and pediatric cancers, but not to preterm birth, stillbirth, or low birth weight. Overall, the findings revealed a lack of consistency in the definition of MVM supplementation, and substantial variability in MVM effectiveness depending on population, age, and health status. These results highlighted the importance of shifting from generalized supplementation approaches to more targeted, personalized nutritional strategies to support healthspan and longevity.

1. Introduction

Multivitamin and mineral (MVM) supplements are widely used, with millions of adults globally incorporating them into their daily routines. In the United States, approximately 40 % of adults report regular use of MVMs, making them one of the most popular dietary supplements on the market (Knippen et al., 2020). Similarly, the Asia-Pacific region has experienced a significant increase in supplement consumption, with the nutraceutical market valued at 74.6 million USD in 2018 and projected to reach 140.2 million USD by 2027 (Singh et al., 2025). These trends reflect a widespread evidence that MVM supplements can help fill nutritional gaps, support immune function, and promote overall health and enhance healthy aging (Hasan et al., 2024; Macpherson et al., 2013). In both preclinical and clinical studies, individual vitamins and minerals showed benefits for age-related mechanisms. Beyond the established role in calcium and phosphate homeostasis and the prevention of sarcopenia, vitamin D is essential for regulating both innate and acquired immune responses (Ghaseminejad-Raeini et al., 2023). Iron deficiency or destructive metabolism is associated age-related mitochondrial dysfunction and brain aging from higher hydroxyl free radicals (Tian et al., 2022). However, despite extensive research on individual micronutrients, comprehensive evidence summarizing the

* Corresponding author at: Faculty of Behavioural and Movement Sciences, Vrije Universiteit Amsterdam, Amsterdam Movement Sciences, Amsterdam, the Netherlands.

E-mail address: a.b.maier@vu.nl (A.B. Maier).

[1] both authors contributed equally

https://doi.org/10.1016/j.arr.2025.102965
Received 4 June 2025; Received in revised form 23 October 2025; Accepted 23 November 2025
Available online 25 November 2025

effects of MVM supplementation on morality and age-related health outcomes remains limited. Given the increasing global life expectancy and the growing interest in preventive healthcare, it is important to evaluate the clinical significance of MVM supplementation, especially its potential role in promoting healthspan and longevity.

While numerous studies, including randomized controlled trials (RCTs) and meta-analyses, have explored the impact of MVM supplementation on cardiovascular health, cancer incidence, and cognitive decline, the results are often mixed and inconsistent (Macpherson et al., 2013). For instance, one meta-analysis reported no effect of MVM supplementation on lowering cardiovascular-related mortality or stroke risk, though prospective cohort data did show potential benefits for lower incidence of coronary heart disease (Kim et al., 2018). Another meta-analysis on infection rate among adults over 65 years old taking MVM supplements found that undernourished older adults supplemented for over six months reap the most benefit (Stephen and Avenell, 2006). These findings highlight that the response to MVM supplementation may vary across populations, and robust clinical evidence supporting their efficacy generally is still lacking.

This review summarizes meta-analyses investigating the effects of MVM supplementation on a wide range of adult health outcomes across various systems, including cardiovascular, cognitive, psychological, immunological, oncological, ophthalmological, musculoskeletal, and reproductive systems, irrespective of disease status.

2. Methods

2.1. Search strategy

A rapid review was conducted to summarize meta-analyses on the health outcomes of MVM supplementation. Two major databases, *EMBASE* and *MEDLINE*, were used for the literature search. Keywords were employed to capture relevant studies, including “multivitamin,” “multimineral,” “multivitamin and mineral,” “multivitamin and multimineral,” and “vitamin and mineral.” The search was limited to studies published in English and published before 14th March 2025. The protocol of this review was registered on PROSPERO (CRD420251023666). This study adopted a rapid review approach to efficiently synthesize a large and heterogeneous body of evidence on MVM supplementation. This review retained key elements of systematic review rigor, including predefined inclusion criteria, protocol registration, and comprehensive database searches in EMBASE and MEDLINE. However, the independent dual screening and third-party conflict resolution were simplified to accelerate completion. The review followed the core principles of the Preferred Reporting Items for Systematic reviews and Meta-Analyses for Scoping Reviews (PRISMA-ScR), adopting them to a rapid-review context.

2.2. Screening process

The article screening followed a two-phase process using the online tool *COVIDENCE*. Titles and abstracts of all retrieved reviews were first screened to identify meta-analyses examining MVM supplementation and its association with health outcomes, including outcomes in offspring of adults supplemented during pregnancy, while excluding those involving MVM supplementation in children and/or adolescents. Full-text screening assessed eligibility based on pre-determined inclusion criteria, as outlined below.

2.3. Inclusion criteria

Meta-analyses were included in this review if they evaluated case-control studies, cohort studies, and interventional studies that provided data from multiple time points. Cross-sectional studies were excluded, as they do not provide temporal information needed to assess effects or associations over time. This inclusion criteria ensured that a comprehensive range of research approaches was considered, capturing both observational and experimental evidence on the effects of MVM supplementation. In cases where more than one meta-analysis was available for a specific health outcome, only the most recently published and most comprehensive meta-analysis was selected for inclusion, provided that the study types were comparable. This ensured the findings incorporated the recent results on MVM supplementation and its association with health outcomes.

2.4. Data extraction and clustering

For each meta-analysis, extracted data included the first author, year of publication, population characteristics (age, sex, and health status where applicable), disease or health outcome studied, number and study design of included studies (e.g., RCTs or observational), total sample size, and reported effect measures.

The eligible meta-analyses were reviewed to extract relevant data and categorized into health outcome groups to facilitate comparison across different physiological systems, including oncology, cardiology, immunology, psychology, ophthalmology, neurology and musculoskeletal, as well as obstetrics and all-cause mortality. The relative risk (RR), odds ratios (OR), mean difference (MD), standardized mean difference (SMD), and weighted mean difference (WMD) were extracted based on the outcome measured.

3. Results

This review includes 19 meta-analyses encompassing data from at least 5535,426 individuals (one meta-analysis did not report participant numbers). This total consists of 333,943 women during pregnancy and 904,947 children born to mothers who took MVM supplements during pregnancy. The included meta-analyses evaluated the effect of MVM supplementation across a broad range of health outcomes: cardiology (2 meta-analyses; 2043,069 individuals [16 prospective cohort studies, 14 RCTs]), oncology (3 meta-analyses; 1792,504 individuals [7 prospective cohort studies, 5 case-control studies, 1 RCT]), neurology (2 meta-analyses; 8403 individuals [13 RCTs]), psychology (1 meta-analysis; 1292 individuals [8 RCTs]), musculoskeletal (1 meta-analysis; 80,148 individuals [8 observational studies]), immunology (2 meta-analyses; 2313 individuals [23 RCTs]), ophthalmology (2 meta-analyses; 277,732 individuals [12 prospective cohort studies, 15 RCTs]), obstetrics (2 meta-analyses; 235,017 pregnancies [9 observational studies, 3 RCTs]), pediatric health (3 meta-analysis; 1003,873 individuals, comprised of 98,926 pregnancies and 904,947 children [50 observational studies, 4 RCTs]), and all-cause mortality (1 meta-analysis; 91,074 individuals [21 RCTs]). RCTs were primarily assessing the effect of MVM on cognitive function, psychological well-being, immune response, blood pressure regulation, and COVID-19 outcomes. Cohort and case-control studies contributed observational data on cancer incidence, cardiovascular events, pregnancy complications, birth outcomes, and pediatric conditions. The characteristics and outcomes of each meta-analysis are summarized in Table 1 and Fig. 1.

3.1. Oncology

3.1.1. Breast cancer

Meta-analysis of eight observational studies (n = 355,080; five prospective cohort studies and three case-control studies) evaluated the association between MVM supplementation and breast cancer incidence (Chan et al., 2011). The study population included cancer-free women aged 20–83 years, with follow-up durations ranging from 3 to 10 years. Pooled analysis revealed no association between MVM intake and the incidence of breast cancer (RR = 0.99; 95 % confidence interval [CI]: 0.60–1.60 for cohort studies; OR = 1.00; 95 % CI: 0.51–1.97 for case-control studies). Thus, observational findings suggest that MVM supplementation is not associated with the risk of breast cancer.

Table 1
Effect of MVM supplementation on health outcomes. Order. System – publication – health outcomes – study design etc.

System	Health outcomes	Study design	Number of studies	Pooled sample size	Population characteristics	Effect size	First author, year, reference
Oncology	Breast cancer incidence	Cohort & case-control	8	355,080	Women aged 20–83 years, with no prior breast cancer history.	Cohort: RR 0.99 (0.60, 1.60) Case-control: OR 1.00 (0.51, 1.97)	(Chan et al., 2011)
	Colorectal cancer incidence	Cohort	7	708,429	Adults aged 16–89 years, including cancer registry populations.	**RR 0.92 (0.87, 0.97)**	(Heine-Bröring et al., 2015)
	Prostate cancer incidence	RCT, cohort & case-control	5	728,995	Men aged 40–71 years, including cancer registry individuals and hospital-based cases.	Cohort, case-control & RCT: OR 1.11(0.95, 1.29)	Stratton et al., 2011 (Stratton and Godwin, 2011)
	Prostate cancer progression and mortality					Cohort: OR 1.10 (0.97, 1.26)	
Cardiology	Cardiovascular mortality	Cohort & RCT	18	2019,862	Adults (mean age: 57.8 years) without prior cardiovascular disease.	Cohort: RR 1.00 (0.97, 1.04)	Kim et al., 2018 (Kim et al., 2018)
	Coronary heart disease mortality					Cohort: RR 1.02 (0.92, 1.13)	
	Stroke mortality					Cohort: RR 0.95 (0.82, 1.09)	
	Coronary heart disease incidence					Cohort: **RR 0.88 (0.79, 0.97)** RCT: RR 0.97 (0.80, 1.19)	
	Stroke incidence					Cohort: RR 0.98 (0.91, 1.05)	
	Systolic blood pressure, mmHg	RCT	12	23,207	Adults aged 21.9–64.7 years, including normotensive and hypertensive individuals, pregnant women, and those with chronic conditions (e.g., obesity).	**WMD −1.31 (-2.48, −0.14)**	Li et al., 2018 (Li et al., 2018)
	Diastolic blood pressure, mmHg					WMD −0.71 (-1.43, 0.00)	
	Hypertension incidence					OR 0.92 (0.80, 1.05)	
Immunology	Episodes of infection	RCT	20	2100	Adults, both nourished and undernourished, healthy and unwell. Included community-dwelling healthy adults, hospitalized patients, and those with chronic conditions.	Older adults (≥65 years): WMD 0.06 (-0.04, 0.16) **Younger adults (<65 years): WMD −1.20 (-2.08, −0.32)**	Stephen et al., 2006 (Stephen and Avenell, 2006)
	COVID-19 ICU admission	RCT	3	213	Hospitalized patients (aged 18–75 years), including those admitted to ICU.	RR 0.94 (0.72,1.21)	Sinopoli et al., 2024 (Sinopoli et al., 2024)
	COVID-19 hospital length of stay, days					WMD −0.20 (-1.08, 0.68)	
	COVID-19 oxygen saturation improvement					WMD 0.04 (-0.02, 0.10)	
	COVID-19 mortality					RR 0.96 (0.73,1.25)	
Psychology	Stress level	RCT	8	1292	Healthy adults, typically aged 18–65 years (some studies reported mean age instead of a range), including highly stressed but healthy individuals.	**SMD 0.35 (0.47, 0.22)**	Long et al., 2013 (Long and Benton, 2013)
	Mild psychiatric symptoms level					**SMD 0.30 (0.43, 0.18)**	
	Anxiety level					**SMD 0.32 (0.48, 0.16)**	
	Depressive symptoms level					SMD 0.20 (0.42, 0.03)	
	Fatigue level					**SMD 0.27 (0.40, 0.15)**	
	Confusion level					**SMD 0.23 (0.38, 0.07)**	
Opthalmology	Visual acuity level	RCT	13	85,321	Adults aged 42–89 years, including those with early or moderate age-related macular degeneration or at risk of late age-related macular progression.	WMD 0.02 (-0.07, 0.01)	Li et al., 2022 (Li et al., 2022)
	Macular degeneration progression					**RR 2.08 (1.41, 3.03)**	
	Nuclear cataract incidence	Cohort & RCT	14	192,411	Adults aged ≥ 40 years, from population-based cohorts and clinic-based studies. Individuals were mostly healthy older adults, with some studies including individuals with early cataracts.	Cohort: **RR 0.73 (0.64, 0.82)** RCT: **RR 0.66 (0.50, 0.88)**	Zhao et al., 2014 (Zhao et al., 2014)
	Cortical cataract incidence					Cohort: **RR 0.81 (0.68, 0.94)**	
	Posterior subscapular					Cohort: RR 0.96 (0.72, 1.20)	

(continued on next page)

Table 1 (*continued*)

System	Health outcomes	Study design	Number of studies	Pooled sample size	Population characteristics	Effect size	First author, year, reference
	cataracts incidence Any type of cataract incidence Cataract surgery needed					Cohort: **RR 0.66 (0.39, 0.93)** Cohort: RR 1.00 (0.92, 1.08)	
Musculoskeletal	Fragility hip fracture incidence	Cohort & case-control	8	80,148	Adults (average age: 69 ± 5.3 years), including hospitalized patients with osteoporotic hip fractures.	**OR 0.49 (0.32, 0.77)**	Beeram et al., 2021 (Beeram et al., 2021)
Neurology	Verbal fluency Immediate free recall Delayed free recall	RCT	10	3200	Adults aged > 18 years, cognitively intact, including those with cardiovascular risk factors or frailty.	SMD 0.06 (-0.05, 0.18) **SMD 0.32 (0.09, 0.56)** SMD −0.14 (-0.43, 0.14)	Grima et al., 2012 (Grima et al., 2012)
	Global cognition Episodic memory	RCT	3	5203	Adults aged ≥ 60 years, without dementia at baseline, including those with cardiovascular risk factors.	**MD 0.07 (SD 0.03, 0.11)** **MD 0.06 (SD 0.03, 0.10)**	Vyas et al., 2024 (Vyas et al., 2024)
Obstetrics	Gestational hypertension	Cohort, Retrospective Secondary Analysis & RCT	6	201,661	Pregnant women, including low-risk and high-risk pregnancies, with variation in baseline hypertension status, folic acid intake, and gestational age at supplementation initiation.	OR 0.91 (0.81, 1.03)	Shim, 2016 (Shim et al., 2016)
	Preeclampsia	Cohort & RCT	4	33,356	Pregnant women, including both low-risk and high-risk groups (e.g., BMI > 25, history of preeclampsia, chronic hypertension, diabetes, or other pre-existing conditions).	Cohort: RR 0.85 (0.69, 1.03) RCT: OR 0.18 (0.03, 0.92), RR ≈ 0.36 (not combined due to clinical heterogeneity)	Christiansen, 2022 (Christiansen et al., 2022)
	Pediatric leukemia Pediatric neuroblastoma Pediatric brain tumour	Case-control	7	NR	Children born to mothers who took MVM before or during pregnancy, including high-risk populations (e.g., advanced maternal age, lower socioeconomic status, pre-existing health conditions).	**OR 0.64 (0.53, 0.78)** **OR 0.53 (0.42, 0.68)** **OR 0.73 (0.60, 0.88)**	Goh, 2007 (Goh et al., 2007)
	Preterm birth Low birthweight Small for gestational age Stillbirth Neural tube defects	Cohort, RCT & case-control	35	98,926	Pregnant women from high-income countries, including both low-risk and high-risk pregnancies (e.g., maternal obesity, pre-existing hypertension, diabetes).	Cohort & RCT: RR 0.84 (0.69, 1.03) Cohort: RR 0.79 (0.45, 1.41) Cohort: **RR 0.77 (0.63, 0.93)** Cohort: RR 0.78 (0.59, 1.03) Cohort & case-control: **RR 0.67 (0.52, 0.87)**	Wolf, 2017 (Wolf et al., 2017)
	Autism spectrum disorder	Cohort & case-control	12	904,947	Children born to mothers who took MVM before or during pregnancy, including high-risk populations (e.g., advanced maternal age, pre-existing metabolic conditions).	RR 0.74 (0.53, 1.04)	Friel, 2021 (Friel et al., 2021)
Mortality	All-cause mortality	RCT	21	91,074	Adults (average age 62 years), including both healthy individuals and those with chronic conditions.	RR 0.98 (0.94, 1.02)	Macpherson, 2013 (Macpherson et al., 2013)

Abbreviations: COVID-19: coronavirus disease-19; ICU: intensive care unit; NR: not reported; MD: mean difference; OR: odds ratio; RCT: randomized controlled trial; RR: relative risk; SMD: standardized mean difference; WMD: weighted mean difference.
Significant values are in bold.

3.1.2. Colorectal cancer

Pooled analysis from seven prospective cohort studies (n = 708,429) found an 8 % lower in colorectal cancer incidence for multivitamin users (RR = 0.92; 95 % CI: 0.87–0.97) (Heine-Bröring et al., 2015). The study population included adults aged 16–89 years old, with some studies focusing on specific populations, such as healthcare professionals or those at higher risk for colorectal cancer, with follow-up periods ranging from 5 to 18 years. The observational findings suggest a protective association of MVM supplementation with colorectal cancer

3.1.3. Prostate cancer

Meta-analysis of five studies (n = 728,995; one RCT, two case-control, and two cohort studies) examined the association between MVM intake and prostate cancer incidence, progression and mortality (Stratton and Godwin, 2011). The pooled population consisted of men aged 40–71 years, drawn from general population cohorts, hospital-based case-control studies, and population-based registries, with varying baseline cancer risk but no pre-existing prostate cancer at enrolment. Follow-up durations ranged from 5 to 18 years. Analysis found no significant association between MVM intake and prostate cancer incidence (OR = 1.11; 95 % CI: 0.95–1.29). Similarly, the one RCT reported no effect (OR = 0.90; 95 % CI: 0.61–1.33), aligning with the overall findings. Two cohort studies further examined the association between MVM use and advanced/metastatic prostate cancer or prostate cancer mortality, with pooled results showing no significant association (OR = 1.10; 95 % CI: 0.97–1.26). Overall, observational and

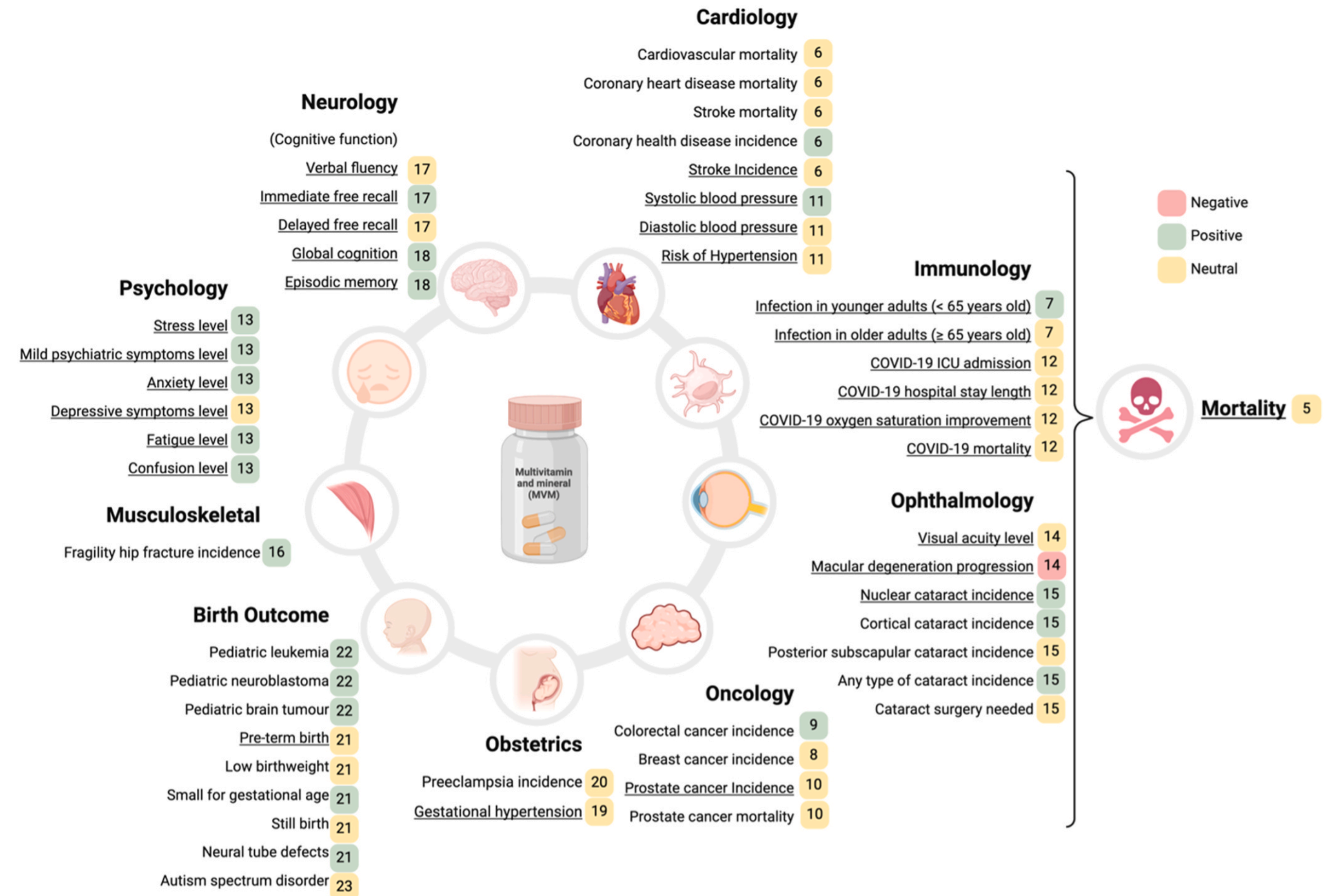

Fig. 1. Effect of MVM supplementation on various health outcomes arranged based on different systems. Health outcomes that are supported by randomized controlled trial (RCT) evidence are underlined.

RCT results indicate that MVM supplementation does not influence prostate cancer incidence, and observational studies found no association with MVM intake on prostate cancer progression or mortality.

3.2. Cardiovascular

3.2.1. Cardiovascular disease outcomes

Meta-analysis of 18 studies (n = 2019,862; 16 cohort studies and two RCTs) examined the association and effect of MVM supplementation on cardiovascular disease outcomes (Kim et al., 2018). The study population included adults (mean age 57.8 years) without pre-existing cardiovascular disease. The follow-up duration ranged from 5 to 19.1 years. The pooled analysis of cohort studies found no significant association between MVM use and cardiovascular disease mortality (RR = 1.00; 95 % CI: 0.97–1.04), coronary heart disease mortality (RR = 1.02; 95 % CI: 0.92–1.13), stroke mortality (RR = 0.95; 95 % CI: 0.82–1.09), or stroke incidence (RR = 0.98; 95 % CI: 0.91–1.05). A lower risk of coronary heart disease incidence was observed (RR = 0.88; 95 % CI: 0.79–0.97), though such association was not observed in RCTs (RR = 0.97; 95 % CI: 0.80–1.19). Overall, observational studies found an association between MVM intake and lower coronary heart disease incidence, while RCT evidence does not support a causal effect.

3.2.2. Blood pressure

A meta-analysis of 12 RCTs (n = 23,207) examined the impact of MVM supplementation on blood pressure (Li et al., 2018). The study population included adults aged 21–65 years old, comprising both normotensive and hypertensive individuals, and those with chronic diseases (e.g., diabetes, cardiovascular conditions), with follow-up durations ranging from 8 weeks to 5 years. The pooled analysis found that MVM supplementation led to a small but significant reduction in systolic blood pressure (SBP) (-1.31 mmHg; 95 % CI: −2.48 to −0.14), but no significant reduction in diastolic blood pressure (DBP) (-0.71 mmHg; 95 % CI: −1.43–0.00). Subgroup analyses founder greater reductions of blood pressure in hypertensive individuals (-7.98 mmHg SBP, 95 % CI: −14.95 to −1.02 mmHg) and those with chronic diseases (-6.29 mmHg SBP, 95 % CI: −11.09, −1.50 mmHg to −2.32 mmHg DBP, 95 % CI: −4.50 to −0.13 mmHg), while effects in healthy normotensive individuals were minimal. Furthermore, MVM supplementation did not affect the risk of developing hypertension in normotensive individuals (OR = 0.92; 95 % CI: 0.80–1.05). These RCT findings suggest that MVM intake lowers blood pressure in individuals with hypertension or chronic disease but has little effect in normotensive individuals and does not prevent hypertension.

3.3. Immunity

3.3.1. Infection

Meta-analysis of 20 RCTs (n = 2100) assessed whether MVM supplementation effectively reduces infection rates in adults across community settings, hospitals, nursing homes, and outpatient clinics (Stephen and Avenell, 2006). The study population included both healthy and unwell individuals, as well as nourished and undernourished groups, with supplementation duration ranging from 8 weeks to 2 years. Pooled analysis found no significant effect of MVM supplementation on infection rates in older adults (≥65 years) (weighted mean difference [WMD] = 0.06; 95 % CI: −0.04–0.16) or the proportion of individuals experiencing at least one infection (RR = 0.98; 95 % CI: 0.86–1.11). Among younger adults (<65 years), MVM supplementation significantly reduced the number of infections (WMD = −1.20; 95 % CI: −2.08 to −0.32), though it did not significantly lower the risk of individuals experiencing at least one infection (RR = 0.81; 95 % CI: 0.65–1.00). In subgroup analysis, undernourished older adults, MVM supplemented for at least 6 months reduced the number of infections (WMD = −0.67; 95 % CI: −1.24 to −0.10). Overall, RCT findings indicate that MVM supplementation does not reduce infection risk in older adults from populations that are a mix of community-dwelling and institutionalized individuals who are typically healthy, but it may lower infection rates in younger adults from populations that are hospitalized, undernourished, and acutely ill, particularly burn patients.

3.3.2. Coronavirus disease-2019 management

Meta-analysis of three RCTs (n = 213) assessed the effects of MVM supplementation in managing coronavirus disease-2019 (COVID-19) (Sinopoli et al., 2024). The study population included hospitalized adults aged 18–75 years, including intensive care unit-admitted patients. Follow-up ranged from 7 to 40 days. Pooled analysis found no significant effect of MVM supplementation on clinical outcomes, including intensive care unit (ICU) admission rates (RR = 0.94; 95 % CI: 0.72–1.21), length of hospital stay (WMD = −0.20 days; 95 % CI: −1.08–0.68), oxygen saturation improvement (WMD = 0.04; 95 % CI: −0.02–0.10), or mortality (RR = 0.96; 95 % CI: 0.73–1.25). One study reported a reduction in inflammatory markers (WMD = −0.67; 95 % CI: −1.24 to −0.10), but this did not translate into improved recovery time or survival. Overall, RCT findings indicate that MVM supplementation does not impact COVID-19 outcomes in hospitalized patients.

3.4. Psychology

A meta-analysis of eight RCTs (n = 1292) examined the impact of MVM supplementation on stress, mild psychiatric symptoms, and mood in healthy adults aged 18–69 years (Long and Benton, 2013). Intervention durations ranged from 28 to 90 days. Pooled analysis found that MVM supplementation significantly reduced perceived stress (SMD = 0.35; 95 % CI: 0.47–0.22), mild psychiatric symptoms (SMD = 0.30; 95 % CI: 0.43–0.18), and anxiety (SMD = 0.32; 95 % CI: 0.48–0.16). Additionally, supplementation reduced fatigue (SMD = 0.27; 95 % CI: 0.40–0.15) and confusion (SMD = 0.23; 95 % CI: 0.38–0.07). No significant effect was observed on depression (SMD = 0.20; 95 % CI: 0.42–-0.03). Further analysis indicated that supplements containing high doses of B vitamins were more effective in improving mood states compared to those with a broader range of micronutrients at lower doses. Overall, RCT findings indicate that MVM supplementation reduces stress, mild psychiatric symptoms, and fatigue in healthy adults, with stronger effects in formulations containing high-dose B vitamins. Further research is required to determine the optimal dosage, ingredient composition, and duration of MVM supplementation.

3.5. Ophthalmology

3.5.1. Macular degeneration

Network meta-analysis of 13 RCTs (n = 85,321) examined the effect of MVM supplementation on age-related macular degeneration progression and visual acuity in adults aged 42–89 years old with varying stages of age-related macular degeneration at baseline, with a follow-up of 9 months to 10 years (Li et al., 2022). While MVM supplementation did not affect visual acuity (WMD = 0.02; 95 % CI: −0.07–0.01), compared to MVM, placebo use was associated with significantly reduced risk age-related macular degeneration progression (RR = 2.08; 95 % CI: 1.41–3.03). Overall, RCT findings suggest that MVM supplementation does not improve visual acuity and increases the risk of progression to late age-related macular degeneration.

3.5.2. Cataracts

Meta-analysis of 14 studies (n = 192,411; 12 prospective cohort studies and two RCTs), examined the effect of MVM supplementation on the risk of age-related cataracts. The study population included adults aged ≥ 40 years, with follow-up durations ranging from 4.8 to 15 years (Zhao et al., 2014). Individuals were generally healthy at baseline, though some studies included individuals with higher cataract risk factors, such as older age (≥65 years), smoking history, high sun exposure, and diabetes or metabolic syndrome. The pooled analysis of cohort studies found that MVM supplementation was associated with a lower risk of nuclear cataracts (RR = 0.73; 95 % CI: 0.64–0.82), cortical cataracts (RR = 0.81; 95 % CI: 0.68–0.94), and any type of cataract (RR = 0.66; 95 % CI: 0.39–0.93). No significant association was found for posterior subcapsular cataracts (RR = 0.96; 95 % CI: 0.72–1.20) or cataract surgery (RR = 1.00; 95 % CI: 0.92–1.08). The two RCTs provided mixed findings, with one trial showing a 34 % reduction in nuclear cataract incidence over 9 years (RR = 0.66; 95 % CI: 0.50–0.88), and another reporting a 43 % reduction in individuals aged 65–74 years (RR = 0.57; 95 % CI: 0.36–0.90). No significant effect was observed in younger individuals aged 45–65 years (RR = 1.28; 95 % CI: 0.76–2.14). Overall, observational findings suggest that MVM supplementation is associated with a lower risk of nuclear, cortical, and overall cataracts, while RCTs provide some evidence supporting a protective effect for nuclear cataracts, particularly in older adults. No significant benefit of MVM supplementation was observed for posterior subcapsular cataracts or cataract surgery.

3.6. Musculoskeletal

Meta-analysis of eight observational studies (n = 80,148) examined the association between MVM supplementation and the risk of fragility hip fractures (Beeram et al., 2021). The study population consisted of older adults (mean age 69 ± 5.3 years) with osteoporotic hip fractures from various population-based cohorts and case-control studies, with a minimum follow-up of one year. Pooled analysis found that MVM supplementation was significantly associated with a lower risk of sustaining a fragility hip fracture (OR = 0.49; 95 % CI: 0.32–0.77). Therefore, observational findings suggest that MVM supplementation is associated with a lower risk of fragility hip fractures.

3.7. Neurology

Meta-analysis of 10 RCTs (n = 3200) investigating the effects of MVM supplementation on cognitive performance in cognitively intact adults over the age of 18 years (Grima et al., 2012). The studies included healthy older individuals, community-dwelling adults, as well as individuals with cardiovascular risk factors or frailty. Supplement duration ranged from 1 to 12 months, with only three RCTs assessing supplementation over a year. Pooled analysis found that MVM supplementation significantly improved immediate free recall memory (SMD = 0.32; 95 % CI: 0.09–0.56). There were no effects for delayed free recall memory (SMD = −0.14; 95 % CI: −0.43–0.14) or verbal fluency (SMD = 0.06; 95 % CI: −0.05–0.18).

Another meta-analysis of three RCT cognitive substudies within the COSMOS trial (n = 5203) with a 2–3 year follow-up examined the effects of MVM supplementation on cognitive performance in adults aged ≥ 60 years without dementia (Vyas et al., 2024). Pooled analysis found significant improvements in global cognition (MD = 0.07; 95 % CI: 0.03–0.11) and episodic memory (MD = 0.06; 95 % CI: 0.03–0.10), with an effect size equivalent to reducing cognitive aging by approximately two years. Overall, RCT evidence indicates that MVM supplementation improves immediate free recall memory in cognitively intact adults and enhances global cognition and episodic memory in older adults without dementia.

3.8. Obstetrics

3.8.1. Gestational hypertension

Meta-analysis of six studies (n = 201,661; two prospective cohort studies, one retrospective cohort, one retrospective secondary analysis, one large population-based cohort, and one RCT) examined the association between MVM supplementation containing folic acid and the risk of gestational hypertension and preeclampsia (Shim et al., 2016). The study population consisted of women with both low- and high-risk pregnancies, and supplementation was initiated at varying gestational ages, ranging from preconception to the second trimester, with follow-up periods extending until delivery. Pooled analysis found no association between MVM supplementation and gestational hypertension or preeclampsia incidence (OR = 0.91; 95 % CI: 0.81–1.03). The single RCT (n = 955) found that MVM supplementation reduced the risk of gestational hypertension (RR = 0.62; 95 % CI: 0.40–0.94). Overall, observational findings suggest no association between MVM supplementation and the risk of gestational hypertension or preeclampsia, while RCT evidence indicates a potential protective effect on gestational hypertension.

3.8.2. Preeclampsia

Meta-analysis of six studies (n = 33,356; four observational studies and two RCTs), examined the association between MVM supplementation and risk of preeclampsia (Christiansen et al., 2022). The study population consisted of women with both low- and high-risk pregnancies with supplementation initiated at varying gestational ages. Supplementation duration varied across studies, with MVM use initiated from preconception to early pregnancy (first trimester) in some studies, while others started supplementation in mid-to-late pregnancy. Follow-up periods extended until delivery to assess preeclampsia incidence. Pooled analysis of observational studies found no significant association between MVM use and preeclampsia risk (RR = 0.85; 95 % CI: 0.69–1.03). The two RCTs (n = 150) reported a protective effect, with one trial finding a significantly lower risk of preeclampsia in the MVM group (OR = 0.18; 95 % CI: 0.03–0.92) and another showing a reduced preeclampsia incidence (RR ≈ 0.36) in high-risk women. due to clinical heterogeneity, their results were not compatible for meta-analysis. Overall, observational findings suggest no association between MVM use and reduced preeclampsia risk, while limited sample size RCTs indicate a protective effect.

3.9. Pregnancy outcomes

3.9.1. Adverse birth outcomes

Meta-analysis of 35 studies (n = 98,926 women; four RCTs and 31 observational studies) examined the effect or association of MVM supplementation before and during pregnancy and adverse birth outcomes (Wolf et al., 2017). The study population included pregnant women from high-income countries, with follow-up until delivery. Pooled findings from both observational and RCT data found no significant effect or association of MVM supplementation on preterm birth (RR = 0.84; 95 % CI: 0.69–1.03). Observational findings for low birth weight (RR = 0.79, 95 % CI: 0.45–1.41) and stillbirth (RR = 0.78; 95 % CI: 0.59–1.03) also found no association, with RCT findings similarly reporting no impact on these outcomes. In contrast, observational findings suggested a protective association between MVM use and small-for-gestational-age births (RR = 0.77; 95 % CI: 0.63–0.93); RCT findings did not support this association. For congenital anomalies, observational findings showed a significant association with MVM use and lower risk of neural tube defects (RR = 0.67; 95 % CI: 0.52–0.87), congenital cardiovascular defects (RR = 0.83; 95 % CI: 0.70–0.98), urinary tract defects (RR = 0.60; 95 % CI: 0.46–0.78), and limb deficiencies (RR = 0.68; 95 % CI: 0.52–0.89), while no association was observed for Trisomy 21 (RR = 0.98; 95 % CI: 0.70–1.37). Overall, observational findings suggest a protective effect of MVM supplementation for certain small-for-gestational-age births and congenital anomalies, whereas pooled findings and RCT evidence do not support an effect on preterm birth, low birth weight, or stillbirth.

3.9.2. Pediatric cancers

Meta-analysis of seven case-control studies (n not reported) explored the association between maternal MVM supplementation during pregnancy and the risk of pediatric cancers (Goh et al., 2007). The study population included children born to mothers who reported taking MVMs before or during pregnancy. The timing and duration of MVM exposure varied, with some mothers starting supplementation before pregnancy and continuing throughout, while others began supplementation only after discovering they were pregnant. Pooled observational findings found a protective association between prenatal MVM use and reduced risk of specific pediatric cancers, namely leukemia (OR = 0.64; 95 % CI: 0.53–0.78), brain tumors (OR = 0.73; 95 % CI: 0.60–0.88) and neuroblastoma (OR = 0.53; 95 % CI: 0.42–0.68). Overall, observational findings suggest maternal MVM supplementation is associated with a lower risk of specific pediatric cancers.

3.9.3. Autism spectrum disorder

Meta-analysis of 12 observational studies (n = 904,947 children, including 8159 autism cases; seven cohort studies and five case-control studies) examined the association between maternal MVM supplementation during pregnancy and the risk of autism spectrum disorder in offspring (Friel et al., 2021). The study population consisted of children born to mothers who took MVM supplements before or during pregnancy, with cohorts including both general and high-risk populations, such as advanced maternal age and pre-existing metabolic conditions. Pooled analysis found no significant association between prenatal MVM supplementation and autism spectrum disorder risk (RR = 0.74; 95 % CI: 0.53–1.04). In high-quality observational studies, a 23 % lower risk was observed (RR = 0.77; 95 % CI: 0.62–0.96), with a similar protective effect noted in prospective studies (RR = 0.69; 95 % CI: 0.48–1.00) and MVM intake during early pregnancy (RR = 0.76; 95 % CI: 0.58–0.99). Overall, some observational findings suggest a protective effect of prenatal MVM use against autism spectrum disorder in offspring, though pooled analysis found no effect.

3.10. All-cause mortality

Meta-analysis of 21 RCTs (n = 91,074, including 8794 deaths) examined the effect of MVM supplementation on all-cause mortality in independently living adults, with an average age of 62 years and mean supplementation duration of 43 months (Macpherson et al., 2013). Pooled analysis found no effect of MVM supplementation on all-cause mortality (RR = 0.98; 95 % CI: 0.94–1.02). In primary prevention trials, there was a non-significant trend toward reduced mortality (RR = 0.94, 95 % CI: 0.89–1.00). MVM supplementation also did not affect mortality due to vascular causes (RR = 1.01; 95 % CI: 0.93–1.09) or cancer (RR = 0.96; 95 % CI: 0.88–1.04). Overall, RCT studies found that MVM supplementation does not affect all-cause mortality, with no benefit across primary or secondary prevention populations.

4. Discussion

The 19 meta-analyses on MVM supplementation show benefits in specific populations: improved cognitive performance in older adults (Vyas et al., 2024), reduced psychological symptoms in stressed younger adults (Long and Benton, 2013), and reduced episodes of infection in younger, undernourished or hospitalized adults (Stephen and Avenell, 2006). Observational studies found an association of MVM use with lower incidence of cataracts in adults over 40 years (Zhao et al., 2014), coronary heart disease in those without prior cardiovascular disease (Kim et al., 2018), colorectal cancer in adults, including those from cancer registries (Heine-Bröring et al., 2015), and hip fractures in adults, including hospitalized patients with osteoporotic hip fractures (Beeram et al., 2021). Among pregnant women, including both low-risk and high-risk groups, MVM use was associated with reduced risks of small-for-gestational-age births (Wolf et al., 2017), pediatric cancers (Goh et al., 2007), and neural tube defects (Wolf et al., 2017). One analysis reported increased progression of age-related macular degeneration in older adults (Li et al., 2022). While these findings suggest targeted benefits, particularly in undernourished or at-risk populations, others meta-analyses found no effect (Chan et al., 2011; Stratton and Godwin, 2011).

MVM supplementation is widely marketed to address nutritional gaps and support overall health, particularly in older adults. The concept of healthy longevity, emphasizing not just lifespan but also years lived in good health, or healthspan, was formally elevated as a global priority in 2021 with the launch of the WHO Decade of Healthy Ageing and increased scientific focus on the healthspan-lifespan gap (Garmany et al., 2021). This underscores the need to delay the onset of chronic diseases and functional decline commonly associated with aging. Although MVM supplementation does not significantly impact mortality rates in populations without evidence of nutrient deficiencies (Kim et al., 2018; Long and Benton, 2013; Macpherson et al., 2013; Vyas et al., 2024), it has shown mixed benefits in individuals with lower nutrient intake or at-risk groups, including lowered disease risks and improvements in specific health markers (Rautiainen et al., 2017; Wang et al., 2015), which is relevant in promoting healthy longevity.

The findings from this review highlight that MVM supplementation may confer health benefits in specific subgroups, and therefore age, sex, health status, dietary intake, and food frequency should be considered when recommending MVM use. Dietary intake is particularly important, as subclinical micronutrient deficiencies, often undetectable without biomarker testing, can compromise long-term health and accelerate aging, even when intake appears adequate (Mahadzir et al., 2025). For instance, long-term MVM supplementation has been associated with lower cardiovascular disease mortality among older women in two cohort studies (Bailey et al., 2015; Mursu et al., 2011), with the strongest association observed among adults aged 60 years and above who had consumed MVM supplements for 15 or more years (Bailey et al., 2014). Additionally, daily MVM supplementation in combination with healthy lifestyle habits has been associated with lower mortality risk among individuals with breast cancer (Kwan et al., 2011). In older adults with hip fractures, mobility limitations can impair meal preparation and self-feeding, leading to inadequate dietary intake and micronutrient deficiencies, for which MVM supplementation has been associated with a lower risk of fragility hip fractures (Beeram et al., 2021). One RCT reported fewer infections and fewer days absent from work among participants, particularly those with type 2 diabetes, taking MVM supplements compared to those receiving a placebo (Barringer et al., 2003). In the Physicians' Health Study II, MVM supplementation found no cognitive benefits in older men overall, but suggested that baseline nutritional status may modify the response to supplementation, with greater benefit in those with lower dietary quality (Grodstein et al., 2013; Rautiainen et al., 2017). Positive findings were reported in the COSMOS-Mind study, where older adults receiving daily MVM supplementation experienced significantly less cognitive decline than those receiving placebo (Sachs et al., 2023; Vyas et al., 2024). MVM supplementation has also been shown in an RCT to reduce stress, physical fatigue, and anxiety in healthy young adults aged 20–50 years (Pipingas et al., 2013).

Four meta-analyses included in this review highlighted key limitations in how MVM supplementation is reported, particularly the lack of standardized definitions and contextual information, such as the number of vitamins and minerals required to qualify as an MVM supplement (Kim et al., 2018; Long and Benton, 2013; Macpherson et al., 2013; Vyas et al., 2024). This definitional ambiguity is a recurring issue in the literature, as there is no universally accepted standard for what constitutes an MVM supplement (Yetley, 2007). This inconsistency can lead to significant discrepancies in outcome interpretation and cross-study comparisons, particularly when MVM supplements with different compositions are treated equally (Blumberg et al., 2018b). For example, in pooled cohort analysis (Loftfield et al., 2024), which concluded that MVM supplementation was not associated with lower mortality risk, MVM intake was assessed via self-reported baseline questionnaires, with no information on the supplement's composition beyond frequency of use. This approach fails to account for variations in active ingredients, potentially conflating heterogeneous products in the analysis.

Furthermore, previous reviews have noted systemic biases in evaluating MVM supplementation research due to the "healthy user effect," whereby MVM users tend to engage in healthier lifestyle behaviors and represent a more health-conscious demographic (Fortmann et al., 2013; Macpherson et al., 2013; O'Connor et al., 2022). Individuals with a history of cancer or other chronic diseases were excluded in some observational studies, further limiting generalizability (Loftfield et al., 2024). Although the authors attempted to control for this bias by including only healthy MVM users and non-users, the selected population may not reflect the individuals most likely to benefit from supplementation, such as those with micronutrient deficiencies or inadequate dietary intake. Consequently, the population examed in observational studies and RCTs may differ substantially. Participants recruited into RCT may be healthier and notionally replete, whereas self-selected supplement users in observational cohorts may include individuals with marginal deficiencies or higher baseline disease risks, who are more likely to experience measurable benefits (Blumberg et al., 2018a; Bouillon et al., 2022).

However, on the other hands, a well-designed RCT may better capture the potential benefits of MVM supplementation. For example, the effect sizes for lowering blood pressure with MVM supplementation were greater in RCTs that recruited hypertensive patients, compared to those including only healthy population (Li et al., 2018). Thus, to maximize the effect of MVM supplementation, future RCTs should target populations with demonstrated or laboratory test confirmed micronutrient insufficiency, or those at elevated risk for conditions mechanistically associated to vitamin or mineral deficiency. Stratification by baseline nutritional status, dietary intake, and genetic or lifestyle factors may help identify subgroups most likely to benefit.

Furthermore, none of the meta-analyses included in this review investigated potential variability in individual responses to MVM supplementation. Exploring such heterogeneity may be valuable, as evidence suggests that the effects of MVM supplementation are sex-dependent (Šelb et al., 2024).

The value of MVM supplementation in optimizing the health and healthspan of relatively healthy individuals remains uncertain. However, as this review demonstrates, the evidence for MVM supplementation to promote healthy longevity remains limited and inconclusive. As the conversation around healthspan continues to evolve, recommendations for supplement use must be grounded in solid scientific evidence rather than driven by consumer perception or commercial messaging.

To enhance the effectiveness of MVM supplementation in promoting healthy longevity, it is essential to identify individuals who are most likely to benefit from supplementation. Nutritional needs vary considerably across individuals due to genetics, age, sex, health status, and lifestyle. While generic dietary guidelines offer population-wide benefits (Blumberg et al., 2018a; Coates et al., 2024), personalized nutrition is essential for optimizing the effectiveness of MVM supplements (Ekmekcioglu, 2020; Šelb et al., 2024). Tailoring supplement regimens based on individual physiological and clinical biomarkers can ensure optimal nutrient absorption and utilization (Niculescu and Lupu, 2011), potentially leading to more meaningful clinical outcomes. Advances in genetic testing, nutrigenomics, wearable health devices, and AI-driven analytics have made it possible to design dynamic, data-driven supplement plans responsive to individual health metrics and goals (Grodstein et al., 2013; Marsman et al., 2018). These personalized strategies are important for more precise and targeted nutrition interventions. They may unlock the full potential of MVM supplementation within the broader framework of extending healthspan and promoting healthy longevity.

5. Conclusion

The effects of MVM supplementation on health outcomes are mixed, with potential benefits primarily observed in specific groups such as older adults, individuals with lower baseline dietary quality, and those with chronic conditions (Barringer et al., 2003; Beeram et al., 2021; Grodstein et al., 2013; Pipingas et al., 2013; Rautiainen et al., 2017; Wang et al., 2015). Inconsistencies across study findings, unclear supplement definitions, heterogeneous formulations, and insufficient attention to individual-level factors such as nutritional status (Bailey et al., 2014, 2015; Blumberg et al., 2018b; Kwan et al., 2011; Loftfield et al., 2024; Macpherson et al., 2013; Mursu et al., 2011; Rautiainen et al., 2017; Wang et al., 2015), highlight the need for more targeted research on the effects of MVM supplementation on health outcomes. Future studies should prioritize well-designed RCTs and personalized approaches that account for individual nutritional status, mobility limitations, and chronic health conditions. Such strategies may better optimize the role of MVM supplementation in supporting healthy longevity.

Role of the funding source

This study is funded by the National University of Singapore and Haleon. The funders had no role in study conceptualization, literature search, data collection, date interpretation, or writing of the manuscript.

CRediT authorship contribution statement

WW, VKW, and MDAM were involved in literature search, data collection, data interpretation, writing and editing. ABM was involved in conceptualization, supervision, review, and editing.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Andrea Britta Maier reports a relationship with Biomarkers of Aging Consortium that includes: board membership and non-financial support. Andrea Britta Maier reports a relationship with Healthy Longevity Medicine Society that includes: board membership and non-financial support. Andrea Britta Maier reports a relationship with Academy for Health and Lifespan Research that includes: board membership. Andrea Britta Maier reports a relationship with Chi Longevity that includes: board membership and non-financial support. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Bailey, R., Fakhouri, T., Park, Y., Dwyer, J., Thomas, P., Dodd, K., Gahche, J., Miller, P., Sempos, C., Murray, D., 2014. Multivitamin-mineral use and cardiovascular disease mortality in the U.S. (370.8). FASEB J. 28, 370–378.

Bailey, R.L., Fakhouri, T.H., Park, Y., Dwyer, J.T., Thomas, P.R., Gahche, J.J., Miller, P. E., Dodd, K.W., Sempos, C.T., Murray, D.M., 2015. Multivitamin-mineral use is associated with reduced risk of cardiovascular disease mortality among women in the United States1, 2, 3, 4. J. Nutr. 145, 572–578.

Barringer, T.A., Kirk, J.K., Santaniello, A.C., Foley, K.L., Michielutte, R., 2003. Effect of a multivitamin and mineral supplement on infection and quality of life. A randomized, double-blind, placebo-controlled trial. Ann. Intern Med. 138, 365–371.

Beeram, I., Mortensen, S.J., Yeritsyan, D., Momenzadeh, K., von Keudell, A., Nazarian, A., 2021. Multivitamins and risk of fragility hip fracture: a systematic review and meta-analysis. Arch. Osteoporos. 16, 29.

Blumberg, J.B., Bailey, R.L., Sesso, H.D., Ulrich, C.M., 2018a. The evolving role of multivitamin/multimineral supplement use among adults in the age of personalized nutrition. Nutrients 10.

Blumberg, J.B., Cena, H., Barr, S.I., Biesalski, H.K., Dagach, R.U., Delaney, B., Frei, B., Moreno González, M.I., Hwalla, N., Lategan-Potgieter, R., McNulty, H., van der Pols, J.C., Winichagoon, P., Li, D., 2018b. The use of multivitamin/multimineral supplements: a modified delphi consensus panel report. Clin. Ther. 40, 640–657.

Bouillon, R., Manousaki, D., Rosen, C., Trajanoska, K., Rivadeneira, F., Richards, J.B., 2022. The health effects of vitamin D supplementation: evidence from human studies. Nat. Rev. Endocrinol. 18, 96–110.

Chan, A.L., Leung, H.W., Wang, S.F., 2011. Multivitamin supplement use and risk of breast cancer: a meta-analysis. Ann. Pharm. 45, 476–484.

Christiansen, C.H., Høgh, S., Rode, L., Schroll, J.B., Hegaard, H.K., Wolf, H.T., 2022. Multivitamin use and risk of preeclampsia: a systematic review and meta-analysis. Acta Obstet. Et. Gynecol. Scand. 101, 1038–1047.

Coates, P.M., Bailey, R.L., Blumberg, J.B., El-Sohemy, A., Floyd, Z.E., Goldenberg, J.Z., Gould Shunney, A., Holscher, H.D., Nkrumah-Elie, Y., Rai, D., Ritz, B.W., Weber, W. J., 2024. The Evolution of Science and Regulation of Dietary Supplements: Past, Present, and Future. J. Nutr.

Ekmekcioglu, C., 2020. Nutrition and longevity – From mechanisms to uncertainties. Crit. Rev. Food Sci. Nutr. 60, 3063–3082.

Fortmann, S.P., Burda, B.U., Senger, C.A., Lin, J.S., Whitlock, E.P., 2013. Vitamin and mineral supplements in the primary prevention of cardiovascular disease and cancer: an updated systematic evidence review for the U.S. Preventive Services Task Force. Ann. Intern. Med. 159, 824–834.

Friel, C., Leyland, A.H., Anderson, J.J., Havdahl, A., Borge, T., Shimonovich, M., Dundas, R., 2021. Prenatal vitamins and the risk of offspring autism spectrum disorder: systematic review and meta-analysis. Nutrients 13, 2558.

Garmany, A., Yamada, S., Terzic, A., 2021. Longevity leap: mind the healthspan gap. NPJ Regen. Med. 6, 57.

Ghaseminejad-Raeini, A., Ghaderi, A., Sharafi, A., Nematollahi-Sani, B., Moossavi, M., Derakhshani, A., Sarab, G.A., 2023. Immunomodulatory actions of vitamin D in various immune-related disorders: a comprehensive review. Front. Immunol. 14, 950465.

Goh, Y., Bollano, E., Einarson, T., Koren, G., 2007. Prenatal multivitamin supplementation and rates of pediatric cancers: a meta-analysis. Clin. Pharmacol. Ther. 81, 685–691.

Grima, N.A., Pase, M.P., Macpherson, H., Pipingas, A., 2012. The effects of multivitamins on cognitive performance: a systematic review and meta-analysis. J. Alzheimers Dis. 29, 561–569.

Grodstein, F., O'Brien, J., Kang, J.H., Dushkes, R., Cook, N.R., Okereke, O., Manson, J.E., Glynn, R.J., Buring, J.E., Gaziano, J.M., Sesso, H.D., 2013. Long-term multivitamin supplementation and cognitive function in Men. Ann. Intern. Med. 159, 806–814.

Hasan, M., Allehdan, S., Alalwan, T., Perna, S., Tayyem, R., 2024. Overview of dietary supplements use: a narrative review. Curr. Nutr. Food Sci. 20, 973–981.

Heine-Bröring, R.C., Winkels, R.M., Renkema, J.M., Kragt, L., van Orten-Luiten, A.C., Tigchelaar, E.F., Chan, D.S., Norat, T., Kampman, E., 2015. Dietary supplement use and colorectal cancer risk: a systematic review and meta-analyses of prospective cohort studies. Int J. Cancer 136, 2388–2401.

Kim, J., Choi, J., Kwon, S.Y., McEvoy, J.W., Blaha, M.J., Blumenthal, R.S., Guallar, E., Zhao, D., Michos, E.D., 2018. Association of multivitamin and mineral supplementation and risk of cardiovascular disease: a systematic review and meta-analysis. Circ. Cardiovasc Qual. Outcomes 11, e004224.

Knippen, K.L., Mahas, R., Van Wasshenova, E., 2020. Outcome expectancies, health information seeking, and cancer beliefs associated with multivitamin/mineral use in a national sample, HINTS-FDA 2015. J. Acad. Nutr. Diet. 120, 1368–1376.

Kwan, M.L., Greenlee, H., Lee, V.S., Castillo, A., Gunderson, E.P., Habel, L.A., Kushi, L.H., Sweeney, C., Tam, E.K., Caan, B.J., 2011. Multivitamin use and breast cancer outcomes in women with early-stage breast cancer: the life after cancer epidemiology study. Breast Cancer Res. Treat. 130, 195–205.

Li, K., Liu, C., Kuang, X., Deng, Q., Zhao, F., Li, D., 2018. Effects of multivitamin and multimineral supplementation on blood pressure: a meta-analysis of 12 randomized controlled trials. Nutrients 10, 1018.

Li, S.-S., Wang, H.-H., Zhang, D., 2022. Efficacy of different nutrients in age-related macular degeneration: A systematic review and network meta-analysis. Seminars in ophthalmology. Taylor & Francis, pp. 515–523.

Loftfield, E., O'Connell, C.P., Abnet, C.C., Graubard, B.I., Liao, L.M., Freeman, L.E.B., Hofmann, J.N., Freedman, N.D., Sinha, R., 2024. Multivitamin use and mortality risk in 3 prospective US Cohorts. JAMA Netw. Open 7 e2418729-e2418729.

Long, S.-J., Benton, D., 2013. Effects of vitamin and mineral supplementation on stress, mild psychiatric symptoms, and mood in nonclinical samples: a meta-analysis. Psychosom. Med. 75, 144–153.

Macpherson, H., Pipingas, A., Pase, M.P., 2013. Multivitamin-multimineral supplementation and mortality: a meta-analysis of randomized controlled trials. Am. J. Clin. Nutr. 97, 437–444.

Mahadzir, M.D.A., Tan, S., Elena, S., Chin, E.M., Garg, V., Mantantzis, K., Peter, S., Maier, A.B., 2025. Towards defining optimal concentrations of micronutrients in adults to optimize health. Mech. Ageing Dev. 225, 112062.

Marsman, D., Belsky, D.W., Gregori, D., Johnson, M.A., Low Dog, T., Meydani, S., Pigat, S., Sadana, R., Shao, A., Griffiths, J.C., 2018. Healthy ageing: the natural consequences of good nutrition—a conference report. Eur. J. Nutr. 57, 15–34.

Mursu, J., Robien, K., Harnack, L.J., Park, K., Jacobs, D.R., 2011. Dietary supplements and mortality rate in older women: the iowa women's health study. Arch. Intern. Med. 171, 1625–1633.

Niculescu, M.D., Lupu, D.S., 2011. Nutritional influence on epigenetics and effects on longevity. Curr. Opin. Clin. Nutr. Metab. Care 14, 35–40.

O'Connor, E.A., Evans, C.V., Ivlev, I., Rushkin, M.C., Thomas, R.G., Martin, A., Lin, J.S., 2022. Vitamin and mineral supplements for the primary prevention of cardiovascular disease and cancer: updated evidence report and systematic review for the US preventive services task force. Jama 327, 2334–2347.
Pipingas, A., Camfield, D.A., Stough, C., Cox, K.H.M., Fogg, E., Tiplady, B., Sarris, J., White, D.J., Sali, A., Wetherell, M.A., Scholey, A.B., 2013. The effects of multivitamin supplementation on mood and general well-being in healthy young adults. A laboratory and at-home mobile phone assessment. Appetite 69, 123–136.
Rautiainen, S., Gaziano, J.M., Christen, W.G., Bubes, V., Kotler, G., Glynn, R.J., Manson, J.E., Buring, J.E., Sesso, H.D., 2017. Effect of baseline nutritional status on long-term multivitamin use and cardiovascular disease risk: a secondary analysis of the physicians' health study II randomized clinical trial. JAMA Cardiol. 2, 617–625.
Sachs, B.C., Williams, B.J., Gaussoin, S.A., Baker, L.D., Manson, J.E., Espeland, M.A., Sesso, H.D., Shumaker, S.A., Rapp, S.R., 2023. Impact of multivitamin-mineral and cocoa extract on incidence of mild cognitive impairment and dementia: Results from the COcoa Supplement and Multivitamin Outcomes Study for the Mind (COSMOS-Mind). Alzheimers Dement 19, 4863–4871.
Šelb, J., Cvetko, F., Deutsch, L., Bedrač, L., Kuščer, E., Maier, A.B., 2024. Personalization matters: the effect of sex in multivitamin-multimineral-based cancer prevention. GeroScience 46, 1351–1356.
Shim, S.-M., Yun, Y.-U., Kim, Y.S., 2016. Folic acid alone or multivitamin containing folic acid intake during pregnancy and the risk of gestational hypertension and preeclampsia through meta-analyses. Obstet. Gynecol. Sci. 59, 110–115.
Singh, N.K., Katiyar, R., Kumari, R., Mishra, B., Mandal, S.K., Kumar, S., 2025. Nutraceutical Regulation In Asian Countries. Formulations, Regulations, and Challenges of Nutraceuticals. Apple Academic Press, pp. 127–152.
Sinopoli, A., Sciurti, A., Isonne, C., Santoro, M.M., Baccolini, V., 2024. The efficacy of multivitamin, Vitamin A, Vitamin B, Vitamin C, and Vitamin D supplements in the prevention and management of COVID-19 and long-COVID: an updated systematic review and meta-analysis of randomized clinical trials. Nutrients 16.
Stephen, A.I., Avenell, A., 2006. A systematic review of multivitamin and multimineral supplementation for infection. J. Hum. Nutr. Diet. 19, 179–190.
Stratton, J., Godwin, M., 2011. The effect of supplemental vitamins and minerals on the development of prostate cancer: a systematic review and meta-analysis. Fam. Pract. 28, 243–252.
Tian, Y., Tian, Y., Yuan, Z., Zeng, Y., Wang, S., Fan, X., Yang, D., Yang, M., 2022. Iron metabolism in aging and age-related diseases. Int. J. Mol. Sci. 23.
Vyas, C.M., Manson, J.E., Sesso, H.D., Cook, N.R., Rist, P.M., Weinberg, A., Moorthy, M. V., Baker, L.D., Espeland, M.A., Yeung, L.K., Brickman, A.M., Okereke, O.I., 2024. Effect of multivitamin-mineral supplementation versus placebo on cognitive function: results from the clinic subcohort of the COcoa Supplement and Multivitamin Outcomes Study (COSMOS) randomized clinical trial and meta-analysis of 3 cognitive studies within COSMOS. Am. J. Clin. Nutr. 119, 692–701.
Wang, S.M., Yin, L.Y., Zhang, Y., Fan, J.H., Chang, I.J., Dawsey, S.M., Taylor, P.R., Abnet, C.C., Qiao, Y.L., 2015. Multivitamin and mineral supplementation is associated with the reduction of fracture risk and hospitalization rate in Chinese adult males: a randomized controlled study. J. Bone Min. Metab. 33, 294–302.
Wolf, H.T., Hegaard, H.K., Huusom, L.D., Pinborg, A.B., 2017. Multivitamin use and adverse birth outcomes in high-income countries: a systematic review and meta-analysis. Am. J. Obstet. Gynecol. 217, 404. e401–404. e430.
Yetley, E.A., 2007. Multivitamin and multimineral dietary supplements: definitions, characterization, bioavailability, and drug interactions. Am. J. Clin. Nutr. 85, 269s–276s.
Zhao, L.Q., Li, L.M., Zhu, H., Epidemiological Evidence-Based Eye Disease Study Research Group, 2014. The effect of multivitamin/mineral supplements on age-related cataracts: a systematic review and meta-analysis. Nutrients 6, 931–949.